Statistical Modelling & Analysis (STAT2004/2904/7004)
Assignment 2

Submit your solutions via Blackboard as a single PDF file with A4 page size.

Due Friday 11 September at 4:00 pm

STAT2004/7004: Complete Exercises 1–4 for a maximum of 40 marks and a total of 10%.
STAT2904: Complete Exercises 1–5 for a maximum of 50 marks and a total of 10%.

1. (10 marks) This question builds on problems from Assignment 1. You may use results obtained in Assignment 1 without re-deriving them.

   In Question 1 of Assignment 1 we saw the Rayleigh distribution, which has probability density function

   $$f(x;\theta) = \frac{x}{\theta}\exp\left(-\frac{x^2}{2\theta}\right), \qquad x > 0,\ \theta > 0.$$

   Let $X_1, \ldots, X_n$ be a random sample from this distribution. It may help to know that

   $$\mathbb{E}_\theta(X^2) = 2\theta \qquad \text{and} \qquad \mathbb{E}_\theta(X^4) = 8\theta^2.$$

   (a) Determine the expected information in the sample by **both**: (i) taking the expectation of the observed information; **and** (ii) calculating the variance of the score function. Verify that the two calculations give the same result. [3 marks]

   (b) Does the MLE of θ attain the Cramér–Rao lower bound? Justify your answer. [1 mark]

   (c) Let θ_0 denote the true parameter value for θ. State the limiting distribution of

   $$\sqrt{n}(\hat{\theta} - \theta_0).$$

   [1 mark]

   In Question 3 of Assignment 1 you wrote code to find the maximum likelihood estimate of the parameters of the Lomax distribution, which has probability density function

   $$f(x;\theta) = \frac{\alpha}{\lambda}\left(1 + \frac{x}{\lambda}\right)^{-(\alpha+1)}, \qquad x > 0, \quad \theta = (\alpha, \lambda) \in (0, \infty)^2.$$

   We assume our data is a realisation of a random sample from this distribution.

   (d) Adapt your code to provide standard errors for the MLE of (α, λ) and apply this to the data in the file `lomax.csv`. [2 marks]

   (e) Assume $\alpha > 1$. Suppose we now want the MLE of mean of the distribution

   $$\mathbb{E}_\theta(X) = \frac{\lambda}{\alpha - 1}.$$

   Adapt your code to compute the MLE of the mean of the distribution and the corresponding standard error. [3 marks]

2. (10 marks) Consider the standard linear regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathsf{N}(\mathbf{0}, \sigma^2\mathbf{I}),$$

where $\mathbf{X}$ is an $n \times p$ matrix with full column rank and $n > p$. Let $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ denote the influence matrix.

(a) Let $\mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$ be the vector of residuals. We are interested in the following estimator of σ

$$\hat{\sigma}_c = c\sum_{i=1}^{n} |e_i|$$

where c may depend on the model matrix $\mathbf{X}$ but not the response $\mathbf{Y}$.

i. Determine c so that $\hat{\sigma}_c$ is an unbiased estimator of σ. You may use the fact that $\mathbb{E}|Z| = \sqrt{2/\pi}$ where $Z \sim \mathsf{N}(0, 1)$. [3 marks]

ii. The residuals are generally not independent. Explain why independence is not required in determining the expectation of $\hat{\sigma}_c$. [1 mark]

(b) Let $\mathbf{A}$ is a $p \times p$ invertible matrix, and consider the reparametrised model with model matrix $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{A}$.

i. Express the ordinary least-squares estimator obtained using model matrix $\tilde{\mathbf{X}}$ in terms of $\hat{\boldsymbol{\beta}}$, the ordinary least squares estimator obtained using model matrix $\mathbf{X}$. [1 mark]

ii. Show that the residuals obtain using $\tilde{\mathbf{X}}$ are identical to those obtained using $\mathbf{X}$. [2 marks]

(c) Suppose that the observed response for observation j is increased by an amount δ, while all other observed responses remain unchanged. Thus, the new response vector is

$$\widetilde{\mathbf{Y}} = \mathbf{Y} + \delta\mathbf{u}_j,$$

where $\mathbf{u}_j$ is the vector with a 1 in position j and zeros elsewhere. Let $\mathbf{e}$ and $\widetilde{\mathbf{e}}$ denote the residual vector obtained using $\mathbf{Y}$ and $\widetilde{\mathbf{Y}}$, respectively.

(i) Determine the change in the residuals $\widetilde{e}_j - e_j$ and $\widetilde{e}_i - e_i$ for $i \neq j$. [2 marks]

(ii) Hence, explain why an unusual response at an observation with high leverage may not produce a large residual. [1 mark]

3. (10 marks) A researcher wishes to investigate the effects of two quantitative covariates, x_1 and x_2, on a response variable Y. The researcher can choose the n design points, $(x_{1,1}, x_{2,1}), \ldots, (x_{1,n}, x_{2,n})$, at which the response is observed.

(a) First, consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_{1,i} + \epsilon_i, \quad \epsilon_i \overset{IID}{\sim} \mathsf{N}(0, \sigma^2).$$

For the following parts (i) and (ii) it may be useful to know

$$\begin{pmatrix} n & \sum_{i=1}^{n} z_i \\ \sum_{i=1}^{n} z_i & \sum_{i=1}^{n} z_i^2 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{n} + \frac{\bar{z}^2}{S_{zz}} & -\frac{\bar{z}}{S_{zz}} \\ -\frac{\bar{z}}{S_{zz}} & \frac{1}{S_{zz}} \end{pmatrix},$$

where $\bar{z} = n^{-1} \sum_{i=1}^{n} z_i$ and $S_{zz} = \sum_{i=1}^{n} (z_i - \bar{z})^2$.

i. Derive the variance of $\hat{\beta}_0$. Hence state a condition on the design points $x_{1,1}, \ldots, x_{1,n}$ under which this variance attains its minimum possible value. [1 mark]

ii. Suppose that for all i, $-1 \leq x_{1,i} \leq 1$ and that n is even. How should the $x_{1,i}$ be chosen to minimise the variance of $\hat{\beta}_1$? Justify your answer. [2 marks]

(b) Now consider the multiple linear regression model

$$Y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \epsilon_i, \quad \epsilon_i \overset{IID}{\sim} \mathsf{N}(0, \sigma^2). \tag{1}$$

Assume that each design point is of the form $(x_{1,i}, 0)$ or $(0, x_{2,i})$, and $\sum_{i=1}^{n} x_{1,i} = \sum_{i=1}^{n} x_{2,i} = 0$. For the model matrix to be of full column rank we also need $\sum_{i=1}^{n} x_{1,i}^2 > 0$ and $\sum_{i=1}^{n} x_{2,i}^2 > 0$.

i. Let $\mathbf{X}$ be the model matrix for this model. Determine $\mathbf{X}^T\mathbf{X}$ and hence show this matrix is diagonal. [1 mark]

ii. Suppose the researcher omits x_2 and fits the reduced model

$$Y_i = \alpha_0 + \alpha_1 x_{1,i} + \epsilon_i, \quad \epsilon_i \overset{IID}{\sim} \mathsf{N}(0, \sigma^2). \tag{2}$$

It is not hard to show that $\hat{\alpha}_0 = \hat{\beta}_0$ and $\hat{\alpha}_1 = \hat{\beta}_1$, where $\hat{\beta}_0$ and $\hat{\beta}_1$ are obtained by fitting the full model (1) . If $\beta_2 \neq 0$, explain how omitting x_2 from the model fit affects the usual estimator of the error variance in the reduced model (2). In particular, determine whether this estimator is unbiased, upward biased, or downward biased. [3 marks]

iii. Derive an expression for the leverage of each observation. You will need separate expressions for observations with design points of the form $(x_{1,i}, 0)$ and those of the form $(0, x_{2,i})$. [2 marks]

iv. The researcher suspects that the covariates may interact and attempts to fit

$$Y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \beta_3 x_{1,i} x_{2,i} + \epsilon_i, \quad \epsilon_i \overset{IID}{\sim} \mathsf{N}(0, \sigma^2).$$

Explain why β_3 cannot be estimated with this data. [1 mark]

4. (10 marks) A distributor of temperature-controlled medical supplies wants to predict the time required to complete each delivery route. Accurate predictions are needed for staff scheduling, vehicle allocation, and maintaining delivery windows. For 60 weekday routes, the following were recorded:

   - `distance`: total route distance in kilometres
   - `delivery_stops`: number of delivery stops
   - `packages`: number of temperature-controlled packages
   - `traffic`: index, with 0 representing free-flowing traffic
   - `completion_time`: minutes from depot departure to return

   The data is stored in the file `Delivery.csv`.

   (a) Fit a multiple linear regression model to the `Delivery.csv` data with `completion_time` as the response and the other variables as covariates. Write down the fitted regression model, including the parameter estimates. [2 marks]

   (b) Compute the variance inflation factors for the fitted model. Identify any evidence of collinearity and explain which relationships among the predictors appear to be producing it. [2 marks]

   (c) Use studentised residuals to identify any observations that you consider to be outliers. State and justify the criterion you use to identify outliers. [2 marks]

   (d) Identify the route with the largest Cook's distance. Refit the model after removing this observation and compare the estimated coefficients with those from the original model. Which estimated coefficient changes most relative to its original standard error? Are the substantive conclusions from the model affected? [3 marks]

   (e) Route 51 has a large leverage. Why isn't the Cook's distance for this observation also large. [1 mark]

Only for STAT2904

5. (10 marks) The total signal accumulated by a detector has both expected signal and signal variance proportional to the exposure time. In suitable units, an exposure time $\theta > 0$ produces a measurement distributed as $\mathsf{N}(\theta, \theta)$. Suppose that $X_1, \ldots, X_n$ is a random sample from this distribution.

   (a) Show that $\bar{X}$ is an unbiased estimator of θ, and determine its variance. [1 mark]

   (b) Calculate the Fisher information about θ in the sample, and hence obtain the Cramér-Rao lower bound for the variance of an unbiased estimator of θ. Does $\bar{X}$ attain this lower bound? Justify your answer. [3 marks]

   (c) Express the observations as an intercept-only linear regression model. Use the Gauss-Markov theorem to state an optimality property of $\bar{X}$. [2 marks]

   (d) The Cramér-Rao calculation shows that $\bar{X}$ does not attain the lower bound, whereas the Gauss-Markov theorem shows that $\bar{X}$ is optimal within a particular class of estimators. Explain why these conclusions do not contradict one another. Your answer should identify the class of estimators considered by each theorem. [2 marks]

   (e) Show that the maximum likelihood estimator of θ is

$$\hat{\theta}_{\text{MLE}} = \frac{-1 + \sqrt{1 + 4\overline{X^2}}}{2}, \qquad \text{where} \qquad \overline{X^2} = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

   Explain why this estimator is not a linear function of the observations, and hence why the Gauss-Markov theorem does not compare it with $\bar{X}$. [2 marks]